

Multiple Choice (10 marks)

1. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
2. Statement 1| The original ResNets and Transformers are feedforward neural networks. Statement 2| The original Transformers use self-attention, but the original ResNet does not.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above
4. What is the rank of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
5. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: As the number of training examples approaches infinity, the model's variance decreases, leading to more stable predictions.
7. True or False: Both Bayesians and frequentists agree on the necessity of using prior distributions in all statistical models.
8. True or False: In Naive Bayes, attributes are statistically independent of one another given the class value.
9. True or False: A Bayesian network with nodes H, U, P, and W requires only 7 independent parameters when no assumptions are made.
10. True or False: High entropy in classification suggests that the partitions are pure and consist of a single class label.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check the given decimal with a precision of 2 by using regex.
12. Write a python function to count the number of rectangles in a circle of radius r.

End of Paper